

Core, Legs, and Aerobic Training

No one is you, and that's your power!

Core, legs, and aerobic training are three things that seem relatively unrelated and, perhaps, even could be viewed as unnecessary for rock climbing. Think again!

The core muscles of your torso are called into play for every climbing movement, and they are especially important in climbing vertical to overhanging terrain. A stronger core will enable you to step higher, hang onto slopers better, position your hips for optimal center-of-mass placement, and create torso tension and torque for hard, powerful movements.

Intimately related to the core muscles are the upper-leg and hip muscles that almost always co-contract with the core muscles. You can prove this to yourself by doing a simple plank or a front lever—notice how the glute muscles of your rear end contract with these exercises. In this way the hip flexors and extensors are active players in all climbing movements. Even more dramatic are deadpoint and lunging moves, which require rapid, forceful recruitment of the entire posterior chain—interestingly, these are muscle groups that few climbers ever target for specific training. I hope this chapter will increase awareness of the importance of the posterior chain in climbing, and hopefully you'll take some of the exercises for a test drive and recognize how they will help you climb with more motor control and power.

The closing sections of this chapter examine the topic of body composition and how it relates to performance in a strength-to-weight ratio sport

such as climbing. We'll also examine the somewhat controversial subject of “running for climbers”—should you or shouldn't you run? Obviously, doing a modest amount of aerobic training can help improve your body composition, but will it directly improve your climbing performance? (Hint: For roped climbers the answer is definitely “yes!”). In this chapter we'll sort this out once and for all, so let's get started!

Core and Posterior Chain Leg Training

In climbing, the core muscles of your torso play a key role in enabling your arms and legs to maximize leverage and transfer torque from hand to foot and vice versa. Furthermore, the core muscles are essential in providing the stability you need to stand up on a small foothold and pull on a small crimp hold. But it's on overhanging routes that the core muscles activate most forcefully—twisting your hips, stiffening up on a long reach, throwing and sticking a deadpoint, and keeping your feet on during a super-steep sequence is not possible without total core involvement.

For the average John Doe, the quest for six-pack abs is what comes to mind when speaking of core training. Of course, most climbers know better—there are many muscles that contribute to core strength and the ability to create the stiffness needed for hard movements. To understand how to better train the many core muscles, it's helpful to dig a little deeper and distinguish between the inner core and outer core.

As a conceptual model you can think of the inner core as a protective cylinder that stiffens to stabilize the spine. This cylinder is composed of the abdominals in the front, the spinal erectors on the back, and the obliques on the sides; the diaphragm sits atop the cylinder and the pelvic floor muscles form the bottom of the cylinder. It's these muscle groups that are conventionally viewed as the "core," and they are the target of common exercises such as crunches and Supermans.

All the muscles of the torso—from shoulders to hips—comprise the outer core. For example, the pectorals, trapezius, rhomboids, latissimus dorsi, quadratus lumborum, gluteals, hip flexors, hamstrings, abductors, and adductors are all outer core muscles that provide torso stiffness and drive macro movements.

Proper core training must target both the muscles of the inner and outer core, and importantly, equally strengthen the anterior and posterior muscles. Climbers who just climb, however, tend to be disproportionately stronger in the anterior core muscles—this is one reason for the common "climbers posture" of rounded shoulders, flexed torso, and a flat lower back. Therefore I believe most climbers can benefit from diligent training of the posterior chain of muscles, as detailed in the section on total core and leg training (page 154).

To help facilitate appropriate exercise selection, I've divided the core exercises into three categories according to the area of training focus: anterior core, posterior core, and total core. Performing two exercises from each of the three categories will comprise one excellent core workout.

Anterior Core Muscles

This first group of exercises target the anterior core—that is, pretty much all the muscles you see between your shoulders and hips when you look in the mirror. While the abdominals are the hallmark of the anterior core, there's much more to effectively training this zone than simply cranking out

a hundred crunches. Give the Windshield Wipers exercise a try—it's uniquely effective and my personal favorite!



ABDOMINAL CRUNCH

Okay, let's get this one out of the way. No, it's not a bad exercise, but it often gets too much focus, and occasionally people don't even do it correctly. Here's how to do crunchers that truly isolate the abdominals. Lie on the floor with your legs bent at about 90 degrees and your feet flat on the floor. Cross your arms over your upper chest or place your hands behind your head (harder), but do not interlace your fingers behind your neck. Now lift your shoulder blades off the floor and exhale as you "crunch" upward. The range of motion is small—lift your shoulder blades off the floor, but do not ascend all the way as you would in doing old-school sit-ups. Continue up and down at a brisk pace that takes about one second per repetition, but don't go so fast that you are bouncing off the floor. Perform as many crunches as possible. Your long-term goal is to be able to do fifty to one hundred crunches in a set. As your conditioning improves further, you can perform a second set after a three-minute rest.



1. Starting position.

2. Top position with shoulder blades off the floor.



HANGING KNEE LIFT

This strenuous exercise targets the lower abdominals and hip flexors in a very climbing-specific way—much like lifting your legs on an overhanging route. Do these hanging from a pull-up bar, the bucket holds of a fingerboard, or a set of Pump Rocks with your palms facing away—be sure to engage your scapular stabilizers and maintain tension in your shoulders throughout. Briskly lift your knees up to chest level, allowing your legs to bend naturally with the motion. Pause for a moment, and then lower your legs slowly until they return to a slightly bent position. Immediately begin the next upward repetition, and continue these knee-lift movements at a steady pace until you can no longer perform the full range of motion. Your long-range goal is fifteen to twenty repetitions, at which time you can add ankle weights to increase resistance.



1. Starting position. 2. Top position.

Rest for at least three minutes before performing a second set.



ONE-ARM ELBOW AND SIDE PLANK

There are dozens of plank variations, but this is one of the best because it trains the obliques and hips as well as the anti-extension capabilities of the abdominals. Begin in the standard elbow plank position with only your toes, elbows, and forearms touching the floor. Now lift one arm and continue to open up fully in order to raise the free hand to a vertical position. Contract your inner and outer core muscles as needed to remain stable, and hold this top position for two seconds. Return to the starting position and lift the other arm, opening up to the side and extending the hand to the ceiling. Again, hold the top position for two seconds. This



**1. Starting position.
2. Alternate raising each arm.**

is a good exercise to train endurance, as some core muscles relax and other contract as you alternate sides. Continue for one to two minutes.



ONE-ARM, ONE-LEG PLANK

This strenuous exercise calls into play almost every muscle from your hands to your feet, and as a result it's remarkably hard and tiring! Assume a push-up position with your torso straight and in line with your feet. Spread your feet shoulder width apart, with your toes in contact with the floor. Keeping your arms, back, and legs straight, lift one foot and the opposite hand off the floor for approximately five seconds. Contract the muscles of your arms, shoulders, core, and legs as needed to maintain balance. Switch foot and arm positions so that your other arm and leg are now supporting your weight. Hold this position for about five seconds. Continue alternating the supporting arm and leg every five seconds. End the exercise after one minute, or earlier if you cannot maintain balance on the single arm and leg. Rest for three minutes and perform a second set.



1. Lift one hand and the opposite foot. 2. Then the other.



TRX "MARINE CORE"

With a name like "marine core," this is surely a must-do exercise! Adjust your TRX sling trainer or rings to hip height when you are in a kneeling

position. Kneel about one to two feet away from the handles and extend straight arms to grip the handles or rings with a palms-down grip. Now stiffen up from knees to shoulders and push your hands forward until your arms and torso form a straight line—hold this position for two seconds, and then return to the starting position. This is a strenuous exercise, so proceed with caution; stop if you can't maintain proper form. The goal is six to twelve (hard) repetitions. Do two or three sets with at least a three-minute rest in between.



1. Starting position. 2. Ending position.



WINDSHIELD WIPERS

Along with marine core and the front lever, this is one of my favorite core exercises. Initially it may take you a few workouts to get the hang of this



Begin with shins pulled up to the bar. 1. Drop to 9 o'clock position. 2. Pivot past the starting position and down to a 3 o'clock position.

exercise—but trust that you'll quickly gain the feel and strength to do it like a champ. Begin hanging palms away from a pull-up bar, and then lift your legs upward until your back is more or less parallel to the ground and your shins are near the level of the bar. It helps to bend your arms slightly and think about “bending the bar” with your hands throughout the entire exercise—this will put your shoulders and scapula in the best position for this exercise. Now lower your legs to the side, then immediately raise them back to the top position and lower to the other side. The side-to-side motion is, of course, like windshield wipers—try to “wipe” from 9 o'clock to 3 o'clock and back again. Continue for six to twelve (hard) repetitions. Do two or three sets with adequate rest in between.



FRONT LEVER

Introduced to climbing by the legendary boulderer John Gill, the front lever is the gold standard of core-muscle strength. It is a very difficult gymnastics move, so expect this exercise to feel hard—or even impossible! Fortunately, you can make it a bit easier by simply bending one leg or having a spotter hold your feet. Begin by hanging straight-armed from a bar or a set of rings (harder). Pull up halfway, then push your hands forward, drop your head backward, and lift your legs. Do all this in a single quick motion while attempting to position your entire body—head to toe—parallel to the ground. Squeeze tightly throughout your shoulders, torso, buttocks, and legs to hold this position for two seconds (if you can). It helps to think about pushing your hands toward your hips, even though you'll be in a stationary position. Lower yourself slowly to the starting position and immediately pull up into a front lever again and hold for two seconds. Perform two to five (hard) total front levers. Rest for a minimum of three minutes before performing a second set.



Bent-leg and straight-leg (harder) front lever.

Safety note: The front lever places a great deal of stress on your shoulders and elbows (just like steep climbing), so it is inappropriate for novice or out-of-shape climbers or anyone with ongoing elbow or shoulder problems.

Posterior Core

Exercises that target the posterior core are missing from many climbers' training programs. Taking a few minutes to do a couple of the following exercise would be a wise thing, as would be doing a couple of the total core and leg exercises described in the next section. Do these twice per week for a few months and you'll improve posterior chain proprioception, motor control, and discover a new ability to extend powerfully on steep climbing terrain.



REVERSE PLANK

As the name implies, this is pretty much the reverse of the standard plank described earlier—it strengthens anti-flexion muscles rather than anti-extension as in the downward-facing plank. You can do this exercise with either your elbows or hands as support—either way, you'll work many of the muscles of the shoulders, upper and low back, hips, and hamstrings. Begin in a sitting position with legs extended, then lean back and support your upper body with your elbows and forearms on the floor. Now lift your hips off the floor by driving your elbows and heels into the floor. Attain a position with your shoulders, hips, and knees all in line. Think about squeezing your glutes and lower back muscles to prevent your hips from sagging downward. Hold the reverse plank for twenty to sixty seconds (hard). Do one or two more sets with three minutes rest in between.



Reverse plank.



SUPERMAN

The Superman is an isolation exercise that targets the commonly weak muscles of the lower back. Lie face down on the floor with your arms extended overhead, your legs straight with pointed toes, and your head in a neutral position. Begin by simultaneously raising one arm and the opposite leg as high as is comfortably possible. Hold the top position for

a moment, and then return to the starting position. Repeat by raising the opposite arm and leg off the floor simultaneously. Again, pause for a moment in the top position before returning to the floor. Continue this alternating exercise motion for a total of twenty repetitions or until you can no longer perform a slow, controlled movement. Rest for three minutes before performing a second set.



1. Raise one arm and leg. 2. Then the other.



BACK BRIDGE

There are a number of different ways to back bridge, but this simple version was a favorite of legendary martial artist Bruce Lee for developing a strong lower back and explosive hip extension. Here's how to do it: Lie on your back with arms out to the side, legs bent about 90 degrees, and feet flat on the floor. Press down simultaneously with your feet and shoulders in order to raise your hips and lower back off the floor as far as possible. Strive to explode upward to the top position, hold there for a second and then return slowly to the starting position. Continue this motion for fifteen to twenty reps. Rest for three minutes before performing a second set.



1. Starting position. 2. Top position.



SIDE HIP RAISE

This surprisingly difficult exercise works the oblique muscles along the side of your torso, as well as the quadratus lumborum, gluteus medius,



1. Start/end position. 2. Lowered position.

abductors, and adductors. Lie on your side on the floor and press up with the floor-side arm straight and supporting your weight so that your body forms a triangle with the floor. Rest the free arm along the other (top) side of your body. Keeping the supporting arm straight, lower your hip until it touches the floor and then immediately raise it back up to the starting position. Repeat this lowering and raising of the hip in a slow, controlled manner for ten to twenty (hard) repetitions. Rest for one minute and then switch sides to perform another set.

Total Core and Posterior Chain Leg Training

Why would any climber want to do leg training? The legs are never the primary cause of failure in rock climbing, and added leg muscle mass would surely be a bad thing. Move on—there's nothing important in this section, right?

Read on and apply, that's my message to you! Unless you campus up every route, you do indeed employ forceful leg movements in climbing. Furthermore, the muscles of the posterior chain—spinal erectors, quadratus lumborum, glutes, and hamstrings, among others—are extremely important in providing the core stiffness necessary for steep, powerful moves. Finally, it's rapid rate of force development throughout the posterior chain that adds explosive power to big deadpoint and lunge moves.

Detailed below are four exercises that will fortify your core from shoulders to hips, strengthen the posterior chain, and, in fact, also train your rotator cuff and scapular stabilizers to turn on quickly. Doing these exercises, therefore, is a win, win, win proposition. Furthermore, if you follow the exercise guidelines exactly, I can almost guarantee you won't gain any unwanted muscle mass—neurological strength gains are a wonderful thing for a strength-to-mass ratio athlete!



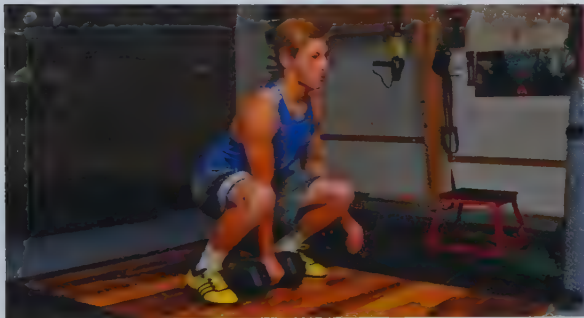
DUMBBELL SNATCH

An important physical capability in sports is being able to quickly turn on (and off) the core muscles. In climbing, sticking a difficult deadpoint or lunge move demands lightning-fast recruitment of your core, from shoulders to hips, at the instant you hit the destination handhold. So in this example your core muscles would be moderately turned on at the launch, relax for a moment while in flight, recruit rapidly and with high force at the catch, and then return to a semi-relaxed state as you gathered yourself after the hard move. While all this happens in just a second or so, it's a critical neuromuscular trait to train both on the bouldering wall and with exercise.

Enter the one-arm dumbbell snatch. While far from a climbing-specific exercise, the dumbbell snatch is beneficial in two ways: (1) It trains explosive triple extension of ankles, knees, and hips (as in a climbing jump move), and (2) It demands a rapid, near-maximal firing of the entire core as you initiate the lift and again when you stick the finish position with the dumbbell straight-arm-locked overhead. Here's how to do it.

Stand with feet just over shoulder width apart and toes pointed about 20 degrees outward. With the dumbbell positioned between your feet, bend equally at your knees and hips to grab onto the dumbbell. At the starting position your lower back should be flat and the working arm hanging straight down from the shoulder; extend the free arm out and backward for balance as needed. The snatch is then executed in one coordinated, continuous, and explosive motion. It begins with knee extension, then hip, knee, and ankle extension simultaneously (the triple extension); the arm remains straight initially and you want to keep the elbow over the dumbbell as long as possible as you jump off the floor. As your shoulders shrug at the end of the jump, allow the dumbbell to launch upward and then drop your body underneath the dumbbell to attain a partial squat position with the dumbbell locked out overhead. Stand up

into the finish position with the dumbbell overhead, shoulders, hips, and feet all in line. Contract your core completely to stick the finish position and hold for one second. Lower the dumbbell and push your hips backward to return to the starting position. This is a technical lift, so begin with a moderate weight (fifteen or twenty-five pounds) and build from there. I suggest pausing for a few seconds between each snatch—let go of the dumbbell, stand tall, and relax for a couple seconds before repeating the lift. Do five to ten repetitions.

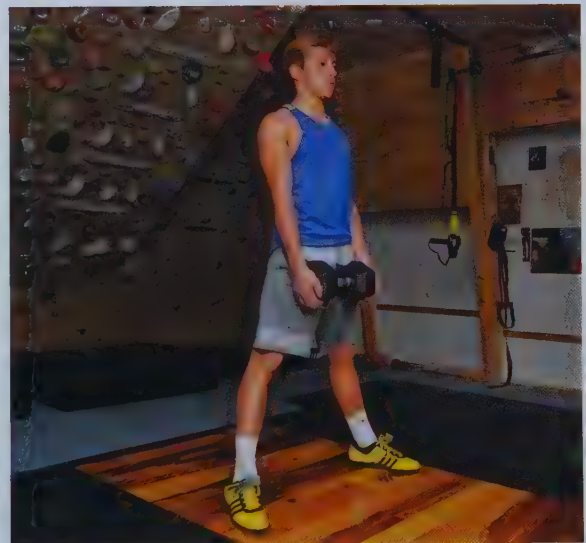


1. Starting position. 2. Ending position.



SUMO DEADLIFT WITH DUMBBELL OR KETTLEBELL

The dumbbell sumo deadlift is an excellent gateway exercise for the more difficult barbell deadlift described next. Both forms of deadlift are excellent for training the posterior chain, core stiffness, the trapezius, and the rotator cuff. I believe that every serious climber should do either this exercise or the barbell deadlift twice per week. The training time investment is small, but the benefits are multiple.



1. Starting position. 2. Ending position.

The starting position is somewhat similar to the dumbbell snatch described above, except that you are using two hands to grasp the ends of the dumbbell (or matching hands if using a kettlebell) and your feet are one and one-half shoulder widths apart with toes angled out about 20 degrees. With flexed knees and hips, a straight back, and arms hanging straight down from the shoulders, tighten your abs and begin the lift by extending the knees. The hips should quickly begin to extend too; however, the knees should extend fully before the hips reach full extension. During the final third of the lift, think about driving your hips forward and then pulling your shoulders back (and scapula together). Finish standing tall, like a soldier at attention, at the top position, but do not hyperextend your back and do not shrug your shoulders. Lower the dumbbell by pushing your rear end back and bending at the hips (maintain a straight back) and then at the knees. Face more or less forward throughout the lift, but with a slight downward gaze near the bottom position.

Train this lift twice per week, favoring a lighter weight and good technique over going super heavy. Many people begin with a fifteen- to thirty-pound dumbbell or kettlebell and build to twice this weight over a year or so. Do two sets of fifteen repetitions. Advanced lifters can do one set of dumbbell sumo deadlifts as a warm-up before doing the barbell deadlift described below.



BARBELL DEADLIFT

It took me until almost age fifty to discover the benefits—and joy!—of deadlifting. As an aging climber with a “bad” lower back, I began a slowly progressive program of deadlifting, stretching, and foam rolling in hopes of erasing decades of abuse. In under two years of training, I went from a fifty-year-old climber with chronic back pain (and in need of back surgery, according to one neurosurgeon) to a rejuvenated, nearly pain-free fifty-two-year-old climber sending 5.13 routes that I

might have struggled on at age thirty! During this two-year ramp-up at deadlift training, I progressed from barely being able to lift 135 pounds to, two years later, deadlifting 300 pounds. But what really matters isn’t the weight I can lift, it’s that my core muscles are now way stronger than ever before, my posterior chain muscles have awoken from a decade-plus of amnesia, and I’m nearly pain-free and climbing much harder than a couple of years ago—talk about a win, win, win! (And, in case you are wondering, I didn’t put on a single pound of body weight.)

The bottom line: Regardless of your age and ability, I strongly recommend some deadlifting if you have access to the weight equipment. No, you shouldn’t be deadlifting like powerlifters (i.e., many heavy sets) or like CrossFitters, who commonly do many high-rep sets with a more moderate load. The protocol I recommend, after a light warm-up set of six reps, is three sets of deadlifts with reps of five, four, and three per set, respectively. Take a three- to five-minute rest between sets, and add ten to twenty pounds with each successive set. The weight used should be heavy enough to make each set difficult, but not so heavy that your technique is compromised. Do this twice per week and you’ll be amazed at how strong you get while gaining little or no muscle mass. Furthermore, I believe you’ll be pleasantly surprised to discover that deadlifting makes you feel very good, and you’ll likely climb better!

The starting position is somewhat similar to the dumbbell sumo deadlift except that your feet should be parallel and a bit less than shoulder width apart. Set up with the bar over the middle of your feet and about one and a half inches from your shin, and then push your rear end back and flex at the hips and knees in order to grab onto the bar with an overhand grip. At the starting position check that your arms are just outside of your legs, your scapula are above the bar, and your knees are over the bar. Now think about squeezing your chest up by engaging the muscles in the mid-back and then

Table 6.1 Deadlifting Protocol for Rock Climbers

Set	Weight (% of 1RM)	Reps/Set	Rest
Warm-up	50–60%	6	3–5 minutes
1	70%	5	3–5 minutes
2	80%	4	3–5 minutes
3	85–90%	2–3	3–5 minutes

This deadlift protocol will maximize neurological strength gains without triggering significant hypertrophy. 1RM (one rep max) is your estimate of the heaviest weight you could deadlift for one repetition.

letting the contraction continue down into the lower back until it is tightened into contraction too. Fix your eyes on the floor at a point twelve to fifteen feet in front of you to maintain a neutral neck position. Before you begin the pull, get the weight back off your toes and onto your mid-foot. Take a deep breath, tighten your abs, and drive your feet



1. Starting position.



2. Ending position.

your feet into the floor—think about dragging the bar up your shins. Focus on driving your hips forward and then pulling your shoulders back (and scapula together) as you approach the top position—your lower back must be kept in extension throughout the lift (not over-extended or rounded). Finish standing tall, like a soldier at attention, but do not hyperextend your back and do not shrug your shoulders. Lower the barbell by pushing your rear end back and bending at the hips (maintain a straight back) and then at the knees. Face more or less forward throughout the lift, but with a slight downward gaze near the bottom position. Pause at the bottom for one second before beginning the next repetition—it's not a deadlift if you bounce the bar off the platform!

Warning: This is a technical lift, and I recommend consulting a coach on your initial forays into the wonderful world of deadlifting. Furthermore, do not deadlift if you have an active back problem; do not deadlift more than twice per week; and do not increase your weights so fast that your form is compromised. But do stretch and foam roll your hamstrings, glutes, and lower back muscles at the end of your workout and before bed on the eve after your deadlifting. I recommend wearing a weight belt when lifting near your 1RM.



BARBELL SQUAT

The squat is another excellent full-body exercise that, like the deadlift, lights up the core muscles in addition to working leg and hip extension. Doing two brief sets (five to eight reps) with moderately heavy weight is a great workout, and it's unlikely to lead to any gain in muscle mass. Learning to squat with proper technique is important—if possible, enlist a coach to help you get started. The lift requires a free weight set and squat rack, so it may not be an option unless you belong to a gym with the necessary equipment; as an alternative, buy a few dumbbells and train with the sumo deadlift described earlier.



1. Starting position. 2. Ending position.

To begin the lift, stand under the racked bar and position it across your mid to upper trapezius, taking a grip wider than shoulder width. Take a deep breath and unrack the bar—carefully back up a step or so to clear the hooks (that were holding the bar). Set your feet about shoulder width and with an outward flare of around 20 degrees. Take another deep breath and contract your inner core, then begin the descent—think about sitting in a chair behind you rather than going straight down. Knees should track over your toes and not collapse inward. Your chest will naturally tilt down as you descend, but strive to keep your weight back on your feet as if sitting down. For a full squat, descend until your thighs are parallel to the floor; beginners may want to stop short of this (three-quarter squat). At the bottom, immediately reverse direction by driving your feet into the floor and extending simultaneously at the knees and hips—think of pushing your head toward the ceiling to quickly regain the top position. Breathe out as you near the top of the lift. Pause for a moment in the top position before taking another deep breath and beginning the next repetition. Be sure to maintain a flat lower back throughout the lift. A final pro tip: Try to bend the bar over your shoulders throughout the lift. While you won't likely bend the bar, this action will maximally recruit your outer core including the latissimus dorsi, pecs, rotator cuff, and trapezius.

Begin squatting with a modest weight, say, around 40 percent of your body weight. Do two sets of five to eight reps (never more), twice per week. Over the course of a year or two, you can work toward squatting your body weight—in my opinion, there's no reason for a climber to train with weights any heavier.

Optimizing Body Composition

Strength-to-weight ratio is the physical metric that most accurately correlates to climbing performance. How small a hold you can pull on, how far you can lunge, and how long you can pull down

before pumping out are all physical capabilities that are influenced by your strength-to-weight ratio. Recognizing this strong correlation, it's not surprising that avid climbers have an insatiable appetite for more strength and power. Interestingly, a climber with less-than-ideal body composition can increase his strength-to-weight ratio most quickly by decreasing weight, not by increasing strength. This is a powerful concept, and for some climbers it points the way to a source of some very fast gains in climbing performance.

A prevailing theme throughout this book is the importance of training smarter, which does not necessarily mean training harder. With regard to the subject of physical training, the smart approach is to pursue near-optimal body composition before delving deeply into a serious strength- and power-training program. Every serious climber should ponder whether or not body composition represents a significant limiting constraint.

Measuring Your Percentage of Body Fat

Most health clubs and some universities have the equipment necessary to measure percentage of body fat, so getting your body fat measured may be just a phone call and a few miles away. A study of athletes in a variety of sports reported that males possess body fat ranging from 4 percent in wrestlers to 8 to 12 percent in runners and 16 percent in football players, with an elite average of below 12 percent (Wilmore 1983). The same study revealed that female athletes possess between 8 and 25 percent body fat, with an elite average of 15 percent. Therefore, a percentage of body fat near these elite average levels (12 percent men, 15 percent women) is a good initial target for most climbers. Given that climbing performance is directly correlated to strength-to-weight ratio, however, your ultimate goal should be a few percentage points lower—perhaps 6 to 8 percent for men and 10 to 12 percent for women. One study (Watts 1993) revealed that some elite male and female climbers possess body fat as low as 4 percent and 9 percent, respectively.

However, extremely low body fat is neither desirable nor advised, since it will adversely affect your energy levels and recovery ability, as well as cause numerous health problems (especially among women).

If you are unable to get a professional body fat measurement, you can always employ the highly economical at-home method—that is, pinch a fold of skin just above your hip! If you can pinch an inch or more (thickness of the fold), you definitely need to drop some body fat. A fold between one-half and one inch thick indicates you may be slightly overweight for a climber. If you pinch less than one-half inch of fat, then your body fat is likely at or below the target averages stated above.

In addition to optimizing your percentage of body fat, you should consider the size and location of the muscles you carry. For instance, it is indisputable that possessing hulking leg muscles is as bad or worse for a climber as carrying a spare tire around the waist (especially since muscle weighs more than fat per unit volume). Since the legs muscles are never the weakest link while climbing, you must be careful how you train the legs and be sure to avoid training in the hypertrophy zone. The same goes for any weight-lifting exercise or practice that produces a bulking-up effect in any other part of the body. Traditional bodybuilding exercises such as biceps curls, heavy-duty leg training, or excessive push-muscle training are of no benefit and likely counterproductive.

Strategies for Optimizing Body Composition

Certainly there are genetic limitations to how much you can change your body composition through diet and exercise. Some people are naturally going to carry a little more body fat; others inherently have a larger frame and bulkier muscles. Still, many novice climbers can improve their body composition significantly in a way that will benefit their climbing. The two key strategies are improved dietary surveillance and increased aerobic training.

Performance nutrition will be discussed in depth in chapter 11, but for now let me state the obvious—reducing body fat is possible only if you create a net calorie deficit over the course of many days and weeks. Crash dieting is unhealthy and dangerous, especially for a serious athlete. Instead strive for, at most, a 500-calorie deficit per day. Over the course of a week, this will add up to a 3,500-calorie deficit and equal the loss of one pound of body fat.

Your daily calorie deficit is best created by a combination of reduced calorie intake and increased energy expenditure. For fat burning, nothing beats aerobic activities such as running, biking, and rowing. Given a healthy back and knees, select running as your activity of choice since for most people it will not result in muscle hypertrophy (growth in size) in the legs. A Concept II rowing machine is another excellent aerobic activity that activates a large amount of muscle mass. Moderate-intensity swimming and biking (on relatively flat terrain) are the next best alternatives. No matter your fat-burning activity, the goal is to perform a minimum of thirty minutes of sustained moderate-intensity aerobic activity at least four days per week.

If your schedule is too busy to accommodate this two-plus hours of aerobics per week, recent research indicates you can get a similar fat-burning effect from shorter, high-intensity interval training (King 2001). For instance, after a three-minute warm-up jog, alternate twenty seconds of high-intensity running (about 90 percent intensity) with forty seconds of jogging for an additional twelve minutes. While this vigorous fifteen minutes of interval training will certainly burn less fat than the thirty minutes of slower steady state running, the metabolism-elevating effects of the sprint intervals will burn more calories in the hours that follow. A final fat-burning tip: A few studies seem to indicate a better fat-burning effect when aerobic activity is performed in a fasting state, as in running first thing in the morning (before breakfast).

Upon reaching your desired percentage of body fat, cut back somewhat on the aerobic training and

Tips for Optimizing Body Composition

1. Use a combination of dietary surveillance and exercise to lower your percentage of body fat.
2. Strive for four thirty-minute (or more) aerobic workouts per week.
3. Adjust calorie intake to produce approximately a 500-calorie-per-day deficit. See chapter 11 for more nutritional tips.
4. Reduce aerobic training activity (and reinvest the training time in climbing-specific exercises) as you approach your optimal percentage of body fat.
5. Do not obsess over body composition—do the best you can given your genetics, and remember that climbing is two-thirds mental and technical. You can climb at a high level despite less-than-ideal body composition.

refocus your training time and effort on climbing-specific activity. Slowly reintroduce more calories incrementally and watch your waistline and body weight. The goal is to find a level of calorie consumption equal to your daily energy use and thus maintain a steady body weight.

In summary, I want to emphasize that while optimizing body composition is an important aspect of training for climbing, it is not everything. Obsessing over body weight and depriving yourself of nutrients—and occasional pleasure foods—is no way to live, and it's certainly not optimal for your climbing performance.

Generalized Aerobic Training

Many climbers shun aerobic training based on the belief that high-end rock climbing is primarily an anaerobic sport. This is understandable given the common experience of pumped-out forearms when failing on a difficult route. The climbing performance

puzzle is extremely complex, however, and it would be a mistake to jump to the simple conclusion.

As detailed in chapter 5, any sustained climbing lasting in excess of two minutes is fueled principally by the aerobic energy pathway rather than the anaerobic. Furthermore, it's the aerobic system that drives the recovery process, no matter if it's at a brief mid-climb rest, between boulders and red-point attempts, or between workouts. Consequently, aerobic training cannot simply be written off as unnecessary or irrelevant. Engaging in a moderate amount of generalized and climbing-specific aerobic training will benefit your climbing performance and accelerate recovery. The focus of this section is generalized aerobic conditioning (local aerobic training of the forearm and pulling muscles will be covered in chapters 8 and 9).

Aerobic Training for Boulderers and Sport Climbers

Excellence at bouldering and cragging requires abundant strength and power, precise technique and efficient movement, and a killer instinct, but much less in the way of stamina. As explained in chapter 5, however, research has shown that climbers who engaged in some generalized aerobic training recovered faster than those who did not. Therefore, doing a modest amount of aerobic training is a smart thing if recovery is important to you.

There's no need for high-volume aerobic training as performed by road racers, bike racers, or triathletes—this would be a foolish investment of training time and would negatively impact your climbing strength and power, due to the catabolic effects of high-volume aerobic training. Furthermore, actual climbing time and climbing-specific training must always take precedence over aerobic training. Therefore, climbing three days per week—and performing some climbing-specific and antagonist strength training—must always take priority over any aerobic training. But if you are getting all your climbing sessions in and still have

time available for additional training, then performing two or three aerobic workouts per week will be beneficial. Let's examine two approaches.



STEADY STATE AEROBIC TRAINING

Tempo runs and Concept 2 rowing are the two best steady state activities for climbers. The goal is to run (or row) at a moderately high intensity that not only stresses the ST muscle fibers, but also begins to recruit some of the FTa (oxidative) fibers. Some people call this a “threshold” workout, since you are exercising near the anaerobic threshold at which lactate and H⁺ ions begin to accumulate. Running at this pace for twenty to thirty minutes not only provides stimulus for favorable cardiovascular adaptations, but it also trains the large exercising muscles (in the legs, hips, back) to better metabolize lactate for energy. The upside, when on a strenuous climb, will be improved processing of lactate and H⁺ released into the bloodstream from the pulling muscles of the upper body. This is one of the reasons climbers who do generalized aerobic training recover faster—and perhaps can persevere longer—on long, pumpy anaerobic climbs.

Sussing out the right intensity of running (or rowing) is the key—you don't want to exercise at too low an intensity (as in a “long slow distance” running pace) or at a high intensity that produces deep fatigue and creates a large acid load on the body. Aim for a rating of perceived exertion of 8 out of 10, or a heart rate around 85 percent of maximum ($220 - \text{age} \times 0.85$). Two twenty- to thirty-minute runs per week are enough to get favorable adaptations. If you desire a third aerobic session each week, I suggest doing an easy “recovery” run with your heart rate nearer 75 percent of maximum.



RUNNING INTERVALS

The most common interval-training program is to run alternating fast and slow laps on a track. Although you can also run intervals on a road or

trail, the ease of setting a goal and gauging distances makes running on a track preferable. For initial training sessions, set out to run two miles (aggregate distance of fast and slow laps). As a rough gauge, your rating of perceived exertion on the fast laps should be about 9 out of 10—hard enough to get significantly winded. Try to hold the fast pace for a complete lap, and then pull back to an easy jog or fast walking for the “slow” lap. Continue alternating fast and slow laps for a total of eight. If road or trail running, alternate between one minute of hard running and two minutes of easy running for about twenty minutes.

This workout will tap significantly into the anaerobic pathway during the fast segments, but the slow segments offer the opportunity to metabolize lactate and buffer H⁺. As described above, aerobic activity that trains the large muscles of the body to increase consumption of blood lactate for energy will hasten the return of homeostasis (in the pumped arm muscles) at rests during (or between) strenuous boulders or roped climbs. The benefits of doing a moderate amount of aerobic training haven't been overlooked by some of the best climbers. Sasha DiGiulian, one of the best endurance sport climbers, and Alex Megos, one of the world's best on-sight climbers, both run regularly as part of their training. Perhaps you should too!

Aerobic Training for Multipitch and Big-Wall Climbers

If your climbing preference is all-day multipitch climbs or big walls, then aerobic training is essential for performing your best in these endeavors. The physical trait you are training for is stamina—the ability to resist fatigue while engaging in sustained or intermittent physical activity for an extended period of time (all day or longer). A regular commitment to long-duration aerobic training will yield local and central adaptations that increase your glycogen stores, improve use of fatty acids for fuel, and strengthen your cardiovascular system. Equally important, however, is the gradual recalibration

of the central governor that results from longer-duration exercise. This sort of “brain training,” a result of pushing to new limits year after year, is the magic behind the training of world-class endurance climbers such as Alex Honnold, Hans Florine, and, of course, Kevin Jorgenson and Tommy Caldwell.

As in other types of conditioning, there are sport-specific and general ways of stamina training. The most specific—and therefore most effective—approach is to train as you will perform. To develop greater stamina for long days at the crags, then, you would train by frequently climbing all-day with the goal of squeezing in as many pitches as possible. Obviously, this approach is not an option for many recreational climbers who only climb outside once per month due to career, school, and family commitments. A more practical training alternative is regular high-volume aerobic activity, such as running and biking, coupled as often as possible with high volumes of climbing at the gym or crag. Let’s take a brief look at these two best ways for training stamina.



ALL-DAY CLIMBING

This is the classic train-as-you-will-perform strategy. If you have the resources nearby, then no training is more specific than chalking up many long days on the rock. You could do this by climbing as many routes as possible from sunrise to sunset at a cragging area or by racing up a Grade IV or V big-wall route in a day. Do this two days per week (or three or four, if you’re already a badass wall rat!) for a few weeks to two months, and you’ll develop amazing climbing stamina! For many climbers, however, it may only be possible to train this way once or twice per month. As a training surrogate, then, put in a few thirty- to fifty-route half-day sessions at the local climbing gym. The routes will need to be fairly moderate (two to four number grades below your limit), and you should employ liberal use of toproping to speed up the process of sending one climb and moving on quickly to the next. To meet the two-day-per-week, long-duration

training goal, you may also need to engage in some generalized aerobic training as well.

HIGH-VOLUME GENERALIZED AEROBIC TRAINING

The goal here is to engage in two to four, forty-five-minute to two-hour aerobic workouts per week that focus on mileage over speed. This could be any combination of running, swimming, cycling, and trail running. As your conditioning improves, consider doubling up sessions one or two days per week. For example, you might go for a long run in the morning and then an hour-long bike ride or trail run in the evening. If you engage in such two-a-day workouts, it is best to take at least a six-hour break between the two. This is clearly an advanced aerobic training program, but it might be just the ticket in the months and weeks leading up to a big-wall or alpine climbing adventure.

Aerobic Training Tips

1. All climbers can benefit from a modest amount of aerobic training. Boulderers and crag climbers will realize accelerated recovery between attempts and ascents, while multipitch and big-wall climbers will gain valuable general endurance for ultralong days and ascents at elevation.
2. Boulderers and sport climbers will benefit the most from twice-per-week interval training or tempo runs near the anaerobic threshold, although aerobic training should never be done at the expense of climbing time.
3. The most effective stamina training for multipitch and big-wall climbers is putting in frequent dawn-to-dusk days on the rock. Alternatively, regular long-distance aerobic activities (running, cycling, hiking, and such) lasting forty-five minutes to two hours will yield substantial gains in aerobic capacity.
4. Two-a-day stamina workouts are effective preparation for single- and multiday wall ascents.

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